ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 4

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B

MARKS: 50 **TIME: 1 hour**

An end-of-year test on Grade 4 Terms 3 and 4: Energy and Change (energy from the Sun and food, food chains, sources of energy, movement and energy in a system, wheels and axles) and Planet Earth and Beyond (the Earth's surface, the Sun, the Moon and rockets).

Written by Academic Central for practice, following the CAPS curriculum for Grade 4. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Write neatly. You may use a pencil for the answers that ask you to draw.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.9 B.

1.1 A food chain always starts with …

A an animal.
B a plant (a producer).
C a carnivore.
D water.

1.2 Which animal is a carnivore?

A Cow B Lion C Goat D Rabbit

1.3 A spring that is pushed together has …

A stored energy.
B light energy.
C sound energy.
D no energy.

1.4 Which of these uses the wind to move?

A A sailing boat B A bicycle C A torch D A candle

1.5 A wheel and axle helps us to …

A move heavy loads more easily.
B make light.
C make food.
D see in the dark.

1.6 Most of the Earth's surface is covered by …

A sand. B water. C ice. D forests.

1.7 The Moon …

A makes its own light.
B is a star.
C moves around the Earth.
D is bigger than the Sun.

1.8 The layer of air around the Earth is called the …

A soil. B atmosphere. C ocean. D desert.

[8]

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number.

2.1 Plants need sunlight to make their own food.

2.2 A herbivore eats only meat.

2.3 People get energy from the food they eat.

2.4 A ball lying still on the ground has movement energy.

2.5 The Sun is much bigger than the Earth.

2.6 There is air to breathe on the Moon.

[6]

QUESTION 3: ENERGY AND FOOD CHAINS

Look at the food chain in a vegetable garden in FIGURE 1 and answer the questions.

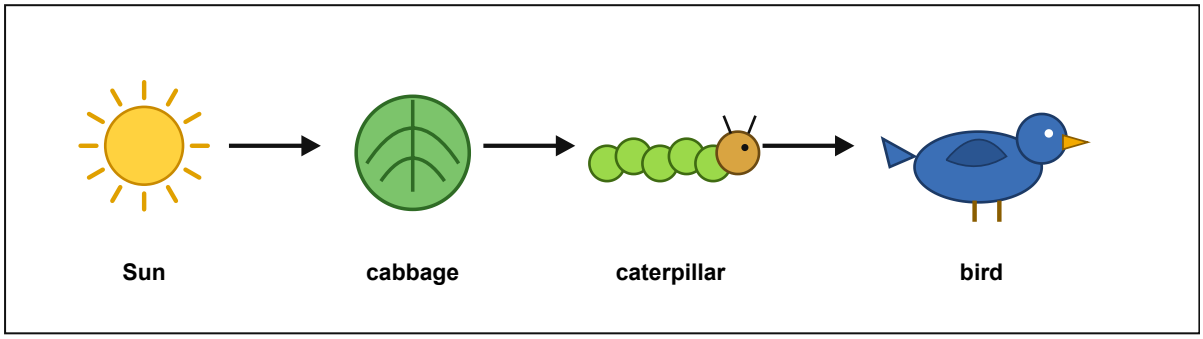


FIGURE 1: A food chain in a vegetable garden

3.1 Name the producer in this food chain. (1)

3.2 Which animal eats the caterpillar? (1)

3.3 Is the caterpillar a herbivore or a carnivore? Give a reason for your answer. (2)

3.4 A farmer kills all the birds in his garden. What will happen to his cabbages? Explain your answer. (2)

3.5 Write your own food chain with THREE living things. Start with a plant and use arrows. (3)

3.6 A lion eats only meat. Explain why a lion still needs plants to stay alive. (2)

3.7 You eat pap made from maize. Explain how energy from the Sun gets into your body. (1)

[12]

QUESTION 4: ENERGY AND MOVEMENT

4.1 What is the source of energy (the input) for each of these?

4.1.1 A windmill pumping water on a farm (1)

4.1.2 A car (1)

4.1.3 A cellphone (1)

4.2 Sipho blew up a balloon and taped it to a straw on a string, as in FIGURE 2. Then he let go of the end of the balloon.

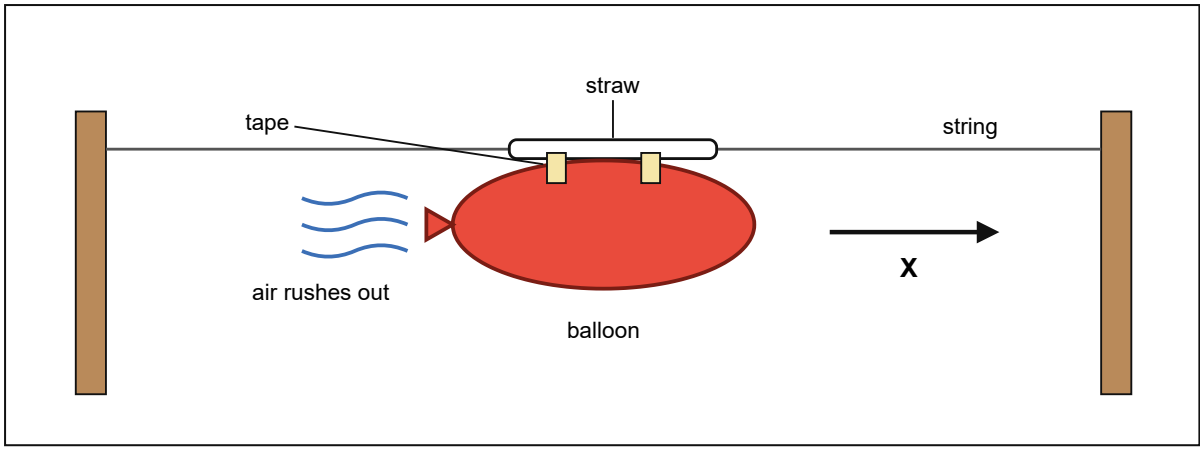


FIGURE 2: Sipho's balloon rocket

4.2.1 Where did the energy to blow up the balloon come from? (1)

4.2.2 Arrow X shows the way the balloon moved. Explain why it moved this way. (2)

4.2.3 A real rocket works in the same way. What does a rocket push out? (1)

4.3 Why is it easier to move a heavy bag of sand in a wheelbarrow than to carry it? (2)

4.4 Name ONE thing we use that has a spring in it. (1)

[10]

QUESTION 5: PLANET EARTH AND BEYOND

5.1 The Earth's surface is made up of THREE parts. Name them. (3)

5.2 Why can't we drink sea water? (1)

5.3 FIGURE 3 shows four shapes of the Moon as we see them from Earth.

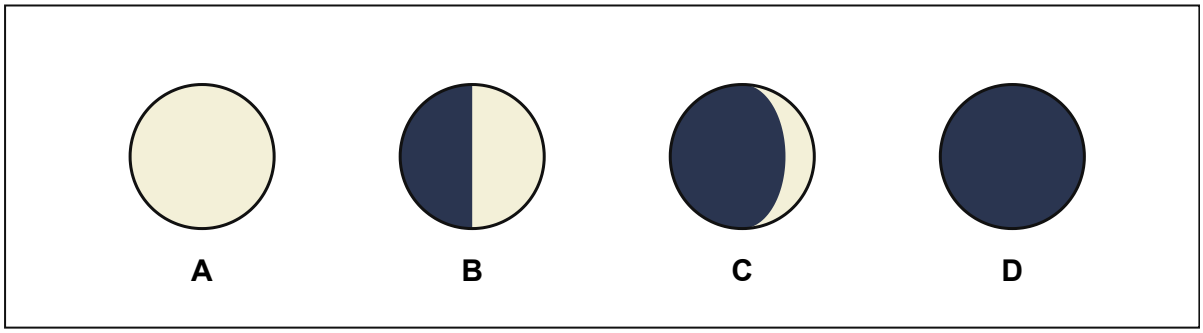


FIGURE 3: Shapes of the Moon

5.3.1 Which letter shows a full moon? (1)

5.3.2 Which letter shows a new moon, when we can hardly see the Moon? (1)

5.3.3 Explain why the Moon seems to change its shape during the month. (2)

5.4 The Earth moves around the Sun. What do we call the path it follows? (1)

5.5 Give TWO ways that the heat of the Sun is useful to people. (2)

5.6 Why should you wear a hat and sunscreen when you play outside in summer? (1)

5.7 Name TWO things astronauts must take with them to the Moon, and say why they need them. (2)

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TOTAL: 50

END OF QUESTION PAPER